

Table 1: Aggregated main results for balanced instances. SA and HG2 are averaged over 10 runs. The best performing method is highlighted in bold.

index	m	n	k	LB	SA	Gurobi	HG2
1	50	480	2	303198	303 590.0	303 590	303 801.2
2	50	480	2	303206	303 678.0	303 678	303 909.2
3	50	480	2	303064	303 596.0	303 596	303 863.6
4	50	480	2	303190	303 586.0	303 586	303 837.2
5	50	480	2	303082	303 606.0	303 606	303 916.4
1	50	1020	2	645672	646 436.0	646 436	646 852.8
2	50	1020	2	645920	646 820.0	646 820	647 039.2
3	50	1020	2	646200	646 756.0	646 756	647 069.6
4	50	1020	2	646008	646 572.0	646 572	646 990.4
5	50	1020	2	645680	646 552.0	646 552	646 828.0
1	50	2040	2	1294544	1 295 452.0	1 295 452	1 295 931.6
2	50	2040	2	1294058	1 294 786.0	1 294 786	1 295 724.0
3	50	2040	2	1294366	1 295 310.0	1 295 310	1 295 681.6
4	50	2040	2	1294132	1 294 980.0	1 294 980	1 295 622.8
5	50	2040	2	1293856	1 295 028.0	1 295 028	1 295 564.8
1	100	480	2	1204148	1 205 544.0	1 205 544	1 206 457.6
2	100	480	2	1203944	1 205 440.0	1 205 440	1 206 231.6
3	100	480	2	1203834	1 205 378.0	1 205 378	1 206 085.2
4	100	480	2	1203938	1 205 363.6	1 205 386	1 206 328.0
5	100	480	2	1203724	1 205 324.0	1 205 324	1 206 255.6
1	100	1020	2	2562896	2 565 507.6	2 565 556	2 566 492.8
2	100	1020	2	2562960	2 565 464.0	2 565 464	2 566 576.8
3	100	1020	2	2562336	2 565 136.0	2 565 136	2 566 172.8
4	100	1020	2	2562960	2 565 455.2	2 565 440	2 566 670.8
5	100	1020	2	2562784	2 565 338.0	2 565 336	2 566 434.4
1	100	2040	2	5131990	5 136 062.0	5 135 986	5 137 682.4
2	100	2040	2	5132328	5 136 156.0	5 136 156	5 137 990.0
3	100	2040	2	5131632	5 135 731.2	5 135 724	5 137 314.8
4	100	2040	2	5131798	5 136 194.8	5 136 190	5 137 829.2
5	100	2040	2	5132624	5 136 345.2	5 136 308	5 137 786.4
1	200	480	2	4803472	4 806 961.6	4 806 938	4 808 896.8
2	200	480	2	4803616	4 806 854.4	4 806 876	4 809 169.6
3	200	480	2	4803677	4 806 804.8	4 806 796	4 809 221.2
4	200	480	2	4803715	4 807 206.4	4 807 118	4 809 192.4
5	200	480	2	4803731	4 807 110.4	4 806 970	4 808 756.4
1	200	1020	2	10217402	10 223 526.8	10 223 344	10 226 421.6
2	200	1020	2	10217154	10 223 658.0	10 223 456	10 226 644.4
3	200	1020	2	10216911	10 223 997.2	10 223 846	10 227 474.0

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Table 1: Aggregated main results for balanced instances.

index	m	n	k	LB	SA	Gurobi	HG2
4	200	1020	2	10217931	10 223 669.2	10 223 584	10 227 077.6
5	200	1020	2	10217462	10 223 831.2	10 223 666	10 227 038.0
1	200	2040	2	20450896	20 460 620.8	20 460 454	20 464 182.4
2	200	2040	2	20451738	20 461 954.8	20 461 634	20 466 354.0
3	200	2040	2	20450765	20 461 043.6	20 460 958	20 466 349.2
4	200	2040	2	20449562	20 460 293.6	20 460 234	20 465 228.8
5	200	2040	2	20450177	20 459 947.6	20 459 750	20 465 499.2
1	300	480	2	10801860	10 807 332.4	10 807 154	10 810 615.6
2	300	480	2	10801946	10 807 298.4	10 807 140	10 810 242.0
3	300	480	2	10801871	10 807 407.2	10 807 196	10 810 182.8
4	300	480	2	10801894	10 807 463.2	10 807 320	10 810 449.2
5	300	480	2	10802017	10 807 458.0	10 807 216	10 810 316.4
1	300	1020	2	22967474	22 978 406.0	22 977 724	22 983 571.2
2	300	1020	2	22967139	22 978 262.0	22 978 168	22 983 768.0
3	300	1020	2	22967234	22 978 316.4	22 978 022	22 983 540.0
4	300	1020	2	22967027	22 978 598.8	22 978 358	22 984 006.8
5	300	1020	2	22966770	22 978 181.2	22 977 852	22 984 399.2
1	300	2040	2	45961393	45 980 006.4	45 979 710	45 987 513.6
2	300	2040	2	45961589	45 980 502.4	45 980 014	45 987 935.2
3	300	2040	2	45959642	45 979 992.0	45 979 290	45 987 574.0
4	300	2040	2	45962020	45 979 969.6	45 979 148	45 986 246.7
5	300	2040	2	45961076	45 979 522.8	45 978 986	45 987 713.6
1	50	480	4	675366	677 017.6	677 088	677 291.0
2	50	480	4	675362	676 924.0	677 050	677 360.6
3	50	480	4	675362	677 014.6	677 066	677 293.4
4	50	480	4	675362	676 987.2	677 020	677 325.8
5	50	480	4	675368	676 954.2	676 950	677 398.6
1	50	1020	4	1435318	1 440 022.6	1 440 156	1 440 497.4
2	50	1020	4	1435299	1 440 039.0	1 440 362	1 440 492.4
3	50	1020	4	1435315	1 440 085.4	1 440 296	1 440 645.8
4	50	1020	4	1435313	1 440 000.0	1 440 232	1 440 606.6
5	50	1020	4	1435313	1 439 946.0	1 440 104	1 440 508.6
1	50	2040	4	2870770	2 882 113.6	2 882 338	2 882 909.0
2	50	2040	4	2870780	2 882 248.0	2 882 308	2 883 107.2
3	50	2040	4	2870770	2 882 088.2	2 882 800	2 882 800.2
4	50	2040	4	2870754	2 882 025.4	2 882 160	2 882 751.0
5	50	2040	4	2870768	2 881 900.0	2 882 288	2 883 113.4
1	100	480	4	2700124	2 702 637.6	2 702 854	2 703 441.8
2	100	480	4	2700116	2 702 616.0	2 702 870	2 703 425.4
3	100	480	4	2700124	2 702 645.6	2 702 984	2 703 442.2

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Table 1: Aggregated main results for balanced instances.

index	m	n	k	LB	SA	Gurobi	HG2
4	100	480	4	2700124	2 702 598.6	2 702 810	2 703 488.8
5	100	480	4	2700124	2 702 594.6	2 702 760	2 703 414.8
1	100	1020	4	5738011	5 746 242.0	5 746 964	5 747 265.6
2	100	1020	4	5737976	5 746 179.4	5 747 454	5 747 345.4
3	100	1020	4	5737977	5 746 194.4	5 746 770	5 747 709.2
4	100	1020	4	5738012	5 746 157.4	5 746 996	5 747 419.8
5	100	1020	4	5737971	5 746 185.2	5 746 896	5 747 452.2
1	100	2040	4	11476037	11 497 449.4	11 498 242	11 499 413.0
2	100	2040	4	11476028	11 497 550.2	11 498 582	11 499 434.2
3	100	2040	4	11476100	11 497 451.6	11 498 944	11 499 679.0
4	100	2040	4	11476048	11 497 400.8	11 498 716	11 499 230.8
5	100	2040	4	11476084	11 497 515.0	11 499 344	11 499 646.0
1	200	480	4	10800000	10 802 937.6	10 803 064	10 804 181.2
2	200	480	4	10800000	10 802 933.2	10 803 406	10 804 229.4
3	200	480	4	10800000	10 802 917.6	10 803 456	10 804 222.6
4	200	480	4	10800000	10 802 922.0	10 803 358	10 804 259.4
5	200	480	4	10800000	10 802 929.6	10 803 712	10 804 271.2
1	200	1020	4	22950097	22 961 903.8	22 963 962	22 964 562.8
2	200	1020	4	22950100	22 961 835.0	22 964 348	22 964 854.2
3	200	1020	4	22950099	22 961 790.8	22 963 656	22 964 427.4
4	200	1020	4	22950102	22 961 803.2	22 963 636	22 964 513.6
5	200	1020	4	22950092	22 961 825.8	22 962 836	22 964 830.4
1	200	2040	4	45900296	45 934 815.4	45 938 588	45 939 577.0
2	200	2040	4	45900272	45 934 618.0	45 939 492	45 939 450.6
3	200	2040	4	45900276	45 935 081.8	45 938 096	45 939 699.6
4	200	2040	4	45900305	45 934 623.2	45 938 704	45 939 500.6
5	200	2040	4	45900271	45 934 929.6	45 938 664	45 939 725.2
1	300	480	4	24300000	24 302 951.1	24 303 304	24 304 433.6
2	300	480	4	24300000	24 302 973.4	24 303 426	24 304 524.8
3	300	480	4	24300000	24 302 976.8	24 303 474	24 304 388.4
4	300	480	4	24300000	24 302 948.2	24 303 420	24 304 280.6
5	300	480	4	24300000	24 302 927.8	24 303 288	24 304 463.4
1	300	1020	4	51637524	51 650 593.6	51 652 966	51 655 002.0
2	300	1020	4	51637530	51 650 713.8	51 651 820	51 654 251.0
3	300	1020	4	51637522	51 650 520.0	51 651 892	51 654 468.0
4	300	1020	4	51637526	51 650 497.8	51 651 496	51 654 467.6
5	300	1020	4	51637526	51 650 654.0	51 652 334	51 655 007.4
1	300	2040	4	103275212	103 317 703.0	103 324 846	103 324 297.4
2	300	2040	4	103275220	103 317 974.4	103 327 686	103 324 902.2
3	300	2040	4	103275217	103 317 597.6	103 327 360	103 324 966.6

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Table 1: Aggregated main results for balanced instances.

index	m	n	k	LB	SA	Gurobi	HG2
4	300	2040	4	103275215	103 318 108.2	103 328 208	103 325 383.6
5	300	2040	4	103275204	103 317 915.0	103 329 640	103 324 962.0
1	50	480	10	972008	972 734.2	972 852	972 932.2
2	50	480	10	972008	972 740.2	972 858	972 912.6
3	50	480	10	972008	972 738.6	972 886	972 966.6
4	50	480	10	972006	972 728.2	972 838	972 935.6
5	50	480	10	972008	972 734.8	972 792	972 912.2
1	50	1020	10	2065562	2 067 774.6	2 068 012	2 068 121.2
2	50	1020	10	2065560	2 067 790.2	2 067 938	2 068 097.6
3	50	1020	10	2065572	2 067 768.2	2 067 992	2 068 151.0
4	50	1020	10	2065576	2 067 769.0	2 067 920	2 068 098.6
5	50	1020	10	2065560	2 067 751.6	2 067 994	2 068 075.6
1	50	2040	10	4131282	4 136 710.2	4 137 094	4 137 203.2
2	50	2040	10	4131280	4 136 661.4	4 137 136	4 137 087.0
3	50	2040	10	4131182	4 136 647.2	4 137 068	4 137 114.4
4	50	2040	10	4131284	4 136 684.6	4 137 004	4 137 164.4
5	50	2040	10	4131166	4 136 696.6	4 137 168	4 137 175.8
1	100	480	10	3888000	3 888 887.6	3 889 008	3 889 183.0
2	100	480	10	3888000	3 888 899.2	3 889 110	3 889 196.8
3	100	480	10	3888000	3 888 911.6	3 889 144	3 889 237.0
4	100	480	10	3888000	3 888 902.2	3 889 089	3 889 218.2
5	100	480	10	3888000	3 888 892.2	3 888 990	3 889 248.2
1	100	1020	10	8262010	8 265 250.0	8 265 872	8 265 908.2
2	100	1020	10	8262010	8 265 266.2	8 265 894	8 265 948.4
3	100	1020	10	8262008	8 265 267.8	8 265 930	8 265 919.0
4	100	1020	10	8262010	8 265 239.4	8 265 754	8 265 881.8
5	100	1020	10	8262010	8 265 235.2	8 265 702	8 265 968.6
1	100	2040	10	16524102	16 533 070.0	16 534 332	16 534 222.4
2	100	2040	10	16524106	16 533 087.2	16 534 130	16 534 252.2
3	100	2040	10	16524104	16 533 155.8	16 533 976	16 534 243.0
4	100	2040	10	16524100	16 533 151.6	16 533 918	16 534 198.2
5	100	2040	10	16524108	16 533 074.4	16 533 926	16 534 183.4
1	200	480	10	15552000	15 552 961.6	15 553 160	15 553 390.0
2	200	480	10	15552000	15 552 969.6	15 553 216	15 553 358.2
3	200	480	10	15552000	15 552 958.6	15 553 552	15 553 377.4
4	200	480	10	15552000	15 552 961.0	15 553 494	15 553 392.2
5	200	480	10	15552000	15 552 960.2	15 553 278	15 553 347.6
1	200	1020	10	33048000	33 052 120.8	33 053 952	33 053 301.0
2	200	1020	10	33048000	33 052 103.4	33 053 990	33 053 250.6
3	200	1020	10	33048000	33 052 101.8	33 053 378	33 053 323.0

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Table 1: Aggregated main results for balanced instances.

index	m	n	k	LB	SA	Gurobi	HG2
4	200	1020	10	33048000	33 052 094.2	33 053 724	33 053 199.6
5	200	1020	10	33048000	33 052 116.8	33 053 698	33 053 330.8
1	200	2040	10	66096012	66 109 291.2	66 112 418	66 111 398.4
2	200	2040	10	66096018	66 109 225.4	66 112 758	66 111 515.8
3	200	2040	10	66096018	66 109 226.0	66 113 274	66 111 665.2
4	200	2040	10	66096016	66 109 189.2	66 113 596	66 111 512.6
5	200	2040	10	66096014	66 109 154.8	66 113 176	66 111 528.0
1	300	480	10	34992000	34 992 978.0	34 994 896	34 993 411.4
2	300	480	10	34992000	34 992 955.2	34 995 400	34 993 400.6
3	300	480	10	34992000	34 992 952.4	34 994 886	34 993 381.4
4	300	480	10	34992000	34 992 959.2	34 995 232	34 993 365.6
5	300	480	10	34992000	34 992 962.4	34 995 240	34 993 346.8
1	300	1020	10	74358000	74 362 386.4	74 365 552	74 363 710.0
2	300	1020	10	74358000	74 362 395.2	74 365 478	74 363 861.2
3	300	1020	10	74358000	74 362 434.6	74 365 182	74 363 765.2
4	300	1020	10	74358000	74 362 446.2	74 365 816	74 363 777.2
5	300	1020	10	74358000	74 362 389.4	74 365 826	74 363 778.4
1	300	2040	10	148716000	148 731 419.6	148 771 016	148 734 916.8
2	300	2040	10	148716000	148 731 456.4	148 770 164	148 734 750.2
3	300	2040	10	148716000	148 731 369.0	148 772 204	148 734 753.6
4	300	2040	10	148716000	148 731 428.0	148 769 438	148 734 507.0
5	300	2040	10	148716000	148 731 381.8	148 770 790	148 734 734.6

Table 2: Aggregated main results for uniform instances. SA and HG2 are averaged over 10 runs. The best performing method is highlighted in bold.

index	m	n	k	LB	SA	Gurobi	HG2
1	50	480	2	280307	280 307.0	280 307	280 307.0
2	50	480	2	283128	283 128.0	283 128	283 128.0
3	50	480	2	286741	286 741.0	286 741	286 741.8
4	50	480	2	283877	283 877.0	283 877	283 884.2
5	50	480	2	286132	286 132.0	286 132	286 140.8
1	50	1020	2	614980	614 980.0	614 980	614 980.0
2	50	1020	2	620179	620 179.0	620 179	620 179.0
3	50	1020	2	611452	611 452.0	611 452	611 452.0
4	50	1020	2	614808	614 808.0	614 808	614 833.6
5	50	1020	2	617663	617 663.0	617 663	617 663.0

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Table 2: Aggregated main results for uniform instances.

index	m	n	k	LB	SA	Gurobi	HG2
1	50	2040	2	1255770	1 255 770.0	1 255 770	1 255 781.2
2	50	2040	2	1249098	1 249 098.0	1 249 098	1 249 098.0
3	50	2040	2	1255319	1 255 319.0	1 255 319	1 255 319.0
4	50	2040	2	1249140	1 249 140.0	1 249 140	1 249 140.0
5	50	2040	2	1261608	1 261 608.0	1 261 608	1 261 634.0
1	100	480	2	1130055	1 130 055.0	1 130 055	1 130 055.0
2	100	480	2	1116714	1 116 714.0	1 116 714	1 116 714.0
3	100	480	2	1126047	1 126 047.0	1 126 047	1 126 047.0
4	100	480	2	1134492	1 134 492.0	1 134 492	1 134 513.6
5	100	480	2	1119719	1 119 719.0	1 119 719	1 119 719.0
1	100	1020	2	2438856	2 438 856.0	2 438 856	2 438 856.0
2	100	1020	2	2444873	2 444 873.0	2 444 873	2 444 873.0
3	100	1020	2	2458524	2 458 524.0	2 458 524	2 458 524.8
4	100	1020	2	2450805	2 450 805.0	2 450 805	2 450 805.0
5	100	1020	2	2453924	2 453 924.0	2 453 924	2 453 924.0
1	100	2040	2	4951606	4 951 606.0	4 951 606	4 951 606.0
2	100	2040	2	4998910	4 998 910.0	4 998 910	4 998 910.0
3	100	2040	2	4974789	4 974 789.0	4 974 789	4 974 789.0
4	100	2040	2	4989491	4 989 491.0	4 989 491	4 989 491.0
5	100	2040	2	4961914	4 961 914.0	4 961 914	4 961 914.0
1	200	480	2	4466723	4 466 723.0	4 466 723	4 466 723.0
2	200	480	2	4521955	4 521 955.0	4 521 955	4 521 955.0
3	200	480	2	4455038	4 455 038.0	4 455 038	4 455 039.6
4	200	480	2	4495362	4 495 362.0	4 495 362	4 495 362.0
5	200	480	2	4483204	4 483 204.0	4 483 204	4 483 204.0
1	200	1020	2	9758812	9 758 812.0	9 758 812	9 758 812.0
2	200	1020	2	9714436	9 714 436.0	9 714 436	9 714 452.8
3	200	1020	2	9747021	9 747 021.0	9 747 021	9 747 021.0
4	200	1020	2	9701528	9 701 528.0	9 701 528	9 701 528.0
5	200	1020	2	9690507	9 690 507.0	9 690 507	9 690 559.0
1	200	2040	2	19825638	19 825 638.0	19 825 638	19 825 649.2
2	200	2040	2	19768108	19 768 108.0	19 768 108	19 768 108.0
3	200	2040	2	19728958	19 728 958.0	19 728 958	19 728 958.0
4	200	2040	2	19780528	19 780 528.0	19 780 528	19 780 528.0
5	200	2040	2	19791397	19 791 397.0	19 791 397	19 791 397.0
1	300	480	2	10087337	10 087 337.0	10 087 337	10 087 337.0
2	300	480	2	10010477	10 010 477.0	10 010 477	10 010 485.8
3	300	480	2	10077428	10 077 428.0	10 077 428	10 077 450.8
4	300	480	2	10037392	10 037 392.0	10 037 392	10 037 392.0
5	300	480	2	10029377	10 029 377.0	10 029 377	10 029 377.0

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Table 2: Aggregated main results for uniform instances.

index	m	n	k	LB	SA	Gurobi	HG2
1	300	1020	2	21935062	21 935 062.0	21 935 062	21 935 062.0
2	300	1020	2	21905248	21 905 248.0	21 905 248	21 905 248.0
3	300	1020	2	21903763	21 903 763.0	21 903 763	21 903 763.0
4	300	1020	2	21875391	21 875 391.0	21 875 391	21 875 396.2
5	300	1020	2	21905694	21 905 694.0	21 905 694	21 905 694.0
1	300	2040	2	44526396	44 526 396.0	44 526 396	44 526 396.0
2	300	2040	2	44486475	44 486 475.0	44 486 475	44 486 475.0
3	300	2040	2	44473526	44 473 526.0	44 473 526	44 473 526.0
4	300	2040	2	44552014	44 552 014.0	44 552 014	44 552 014.0
5	300	2040	2	44452379	44 452 379.0	44 452 379	44 452 379.0
1	50	480	4	640602	640 602.0	640 602	640 602.0
2	50	480	4	639087	639 087.0	639 087	639 087.3
3	50	480	4	635754	635 754.0	635 754	635 754.0
4	50	480	4	635807	635 807.0	635 807	635 807.4
5	50	480	4	636516	636 516.0	636 516	636 516.0
1	50	1020	4	1383332	1 383 332.0	1 383 332	1 383 332.0
2	50	1020	4	1381588	1 381 588.0	1 381 588	1 381 588.0
3	50	1020	4	1386879	1 386 879.0	1 386 879	1 386 879.0
4	50	1020	4	1383112	1 383 112.0	1 383 112	1 383 112.0
5	50	1020	4	1387119	1 387 119.0	1 387 119	1 387 119.0
1	50	2040	4	2795020	2 795 020.0	2 795 020	2 795 020.0
2	50	2040	4	2797936	2 797 936.0	2 797 936	2 797 941.8
3	50	2040	4	2798413	2 798 413.0	2 798 413	2 798 413.0
4	50	2040	4	2799170	2 799 170.0	2 799 170	2 799 170.0
5	50	2040	4	2795753	2 795 753.0	2 795 753	2 795 753.0
1	100	480	4	2539395	2 539 395.0	2 539 395	2 539 395.0
2	100	480	4	2529075	2 529 075.0	2 529 075	2 529 075.0
3	100	480	4	2543218	2 543 218.0	2 543 218	2 543 218.0
4	100	480	4	2539598	2 539 598.0	2 539 598	2 539 611.8
5	100	480	4	2548866	2 548 866.0	2 548 866	2 548 866.0
1	100	1020	4	5509604	5 509 604.0	5 509 604	5 509 604.0
2	100	1020	4	5500449	5 500 449.0	5 500 449	5 500 449.0
3	100	1020	4	5497742	5 497 742.0	5 497 742	5 497 742.0
4	100	1020	4	5505239	5 505 239.0	5 505 239	5 505 239.0
5	100	1020	4	5496108	5 496 108.0	5 496 108	5 496 108.0
1	100	2040	4	11143540	11 143 540.0	11 143 540	11 143 540.0
2	100	2040	4	11137967	11 137 967.0	11 137 967	11 137 967.0
3	100	2040	4	11168575	11 168 575.0	11 168 575	11 168 629.2
4	100	2040	4	11153078	11 153 078.0	11 153 078	11 153 078.0
5	100	2040	4	11168433	11 168 433.0	11 168 433	11 168 445.0

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Table 2: Aggregated main results for uniform instances.

index	m	n	k		LB	SA	Gurobi	HG2
1	200	480	4	10083856	10 083 856.0	10 083 856	10 083 857.2	
2	200	480	4	10158855	10 158 855.0	10 158 855	10 158 855.0	
3	200	480	4	10129398	10 129 398.0	10 129 398	10 129 398.0	
4	200	480	4	10144190	10 144 190.0	10 144 190	10 144 190.0	
5	200	480	4	10170045	10 170 045.0	10 170 045	10 170 054.1	
1	200	1020	4	22068283	22 068 283.0	22 068 283	22 068 283.0	
2	200	1020	4	22060241	22 060 241.0	22 060 241	22 060 253.0	
3	200	1020	4	22022137	22 022 137.0	22 022 137	22 022 137.0	
4	200	1020	4	22058060	22 058 060.0	22 058 060	22 058 060.0	
5	200	1020	4	22007183	22 007 183.0	22 007 183	22 007 183.0	
1	200	2040	4	44693122	44 693 122.0	44 693 122	44 693 122.0	
2	200	2040	4	44632461	44 632 461.0	44 632 461	44 632 461.0	
3	200	2040	4	44600879	44 600 879.0	44 600 879	44 600 881.8	
4	200	2040	4	44685560	44 685 560.0	44 685 560	44 685 560.0	
5	200	2040	4	44589304	44 589 304.0	44 589 304	44 589 304.0	
1	300	480	4	22858710	22 858 710.0	22 858 710	22 858 710.0	
2	300	480	4	22792963	22 792 963.0	22 792 963	22 792 963.0	
3	300	480	4	22826592	22 826 592.0	22 826 592	22 826 592.0	
4	300	480	4	22832475	22 832 475.0	22 832 475	22 832 475.0	
5	300	480	4	22799470	22 799 470.0	22 799 470	22 799 470.0	
1	300	1020	4	49485699	49 485 699.0	49 485 699	49 485 699.0	
2	300	1020	4	49420108	49 420 108.0	49 420 108	49 420 108.0	
3	300	1020	4	49478997	49 478 997.0	49 478 997	49 478 997.0	
4	300	1020	4	49376051	49 376 051.0	49 376 051	49 376 051.0	
5	300	1020	4	49488119	49 488 119.0	49 488 119	49 488 119.0	
1	300	2040	4	100285925	100 285 925.0	100 285 925	100 285 925.0	
2	300	2040	4	100356790	100 356 790.0	100 356 790	100 356 790.0	
3	300	2040	4	100302686	100 302 686.0	100 302 686	100 302 686.0	
4	300	2040	4	100227503	100 227 503.0	100 227 503	100 227 503.0	
5	300	2040	4	100236700	100 236 700.0	100 236 700	100 236 704.5	
1	50	480	10	926804	926 804.0	926 804	926 804.0	
2	50	480	10	927657	927 657.0	927 657	927 657.0	
3	50	480	10	925573	925 573.0	925 573	925 573.0	
4	50	480	10	924941	924 941.0	924 941	924 941.0	
5	50	480	10	924004	924 004.0	924 004	924 004.0	
1	50	1020	10	2000640	2 000 640.0	2 000 640	2 000 640.0	
2	50	1020	10	2003480	2 003 480.0	2 003 480	2 003 480.0	
3	50	1020	10	2002403	2 002 403.0	2 002 403	2 002 403.0	
4	50	1020	10	1998853	1 998 853.0	1 998 853	1 998 853.0	
5	50	1020	10	1997027	1 997 027.0	1 997 027	1 997 027.0	

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Table 2: Aggregated main results for uniform instances.

index	m	n	k	LB	SA	Gurobi	HG2
1	50	2040	10	4041566	4 041 566.0	4 041 566	4 041 566.0
2	50	2040	10	4040638	4 040 638.0	4 040 638	4 040 638.0
3	50	2040	10	4031798	4 031 798.0	4 031 798	4 031 798.0
4	50	2040	10	4046469	4 046 469.0	4 046 469	4 046 469.0
5	50	2040	10	4051439	4 051 439.0	4 051 439	4 051 439.0
1	100	480	10	3702142	3 702 142.0	3 702 142	3 702 142.0
2	100	480	10	3692614	3 692 614.0	3 692 614	3 692 614.0
3	100	480	10	3696598	3 696 598.0	3 696 598	3 696 598.0
4	100	480	10	3699105	3 699 105.0	3 699 105	3 699 105.0
5	100	480	10	3703043	3 703 043.0	3 703 043	3 703 045.5
1	100	1020	10	7988713	7 988 713.0	7 988 713	7 988 713.0
2	100	1020	10	7985011	7 985 011.0	7 985 011	7 985 011.0
3	100	1020	10	7979221	7 979 221.0	7 979 221	7 979 221.0
4	100	1020	10	7993719	7 993 719.0	7 993 719	7 993 719.0
5	100	1020	10	7982658	7 982 658.0	7 982 658	7 982 658.0
1	100	2040	10	16163664	16 163 664.0	16 163 664	16 163 664.0
2	100	2040	10	16129196	16 129 196.0	16 129 196	16 129 196.0
3	100	2040	10	16144148	16 144 148.0	16 144 148	16 144 148.0
4	100	2040	10	16153803	16 153 803.0	16 153 803	16 153 803.0
5	100	2040	10	16133801	16 133 801.0	16 133 801	16 133 801.0
1	200	480	10	14796887	14 796 887.0	14 796 887	14 796 889.2
2	200	480	10	14798454	14 798 454.0	14 798 454	14 798 464.8
3	200	480	10	14775254	14 775 254.0	14 775 254	14 775 254.0
4	200	480	10	14793479	14 793 479.0	14 793 479	14 793 480.4
5	200	480	10	14792851	14 792 851.0	14 792 851	14 792 851.0
1	200	1020	10	31900469	31 900 469.0	31 900 469	31 900 469.0
2	200	1020	10	31964111	31 964 111.0	31 964 111	31 964 111.4
3	200	1020	10	31968514	31 968 514.0	31 968 514	31 968 514.0
4	200	1020	10	31946395	31 946 395.0	31 946 395	31 946 395.0
5	200	1020	10	31949064	31 949 064.0	31 949 064	31 949 064.0
1	200	2040	10	64508050	64 508 050.0	64 508 050	64 508 050.0
2	200	2040	10	64548312	64 548 312.0	64 548 312	64 548 314.3
3	200	2040	10	64558958	64 558 958.0	64 558 958	64 558 958.0
4	200	2040	10	64568971	64 568 971.0	64 568 971	64 568 971.0
5	200	2040	10	64504580	64 504 580.0	64 504 580	64 504 580.0
1	300	480	10	33319281	33 319 281.0	33 319 281	33 319 290.2
2	300	480	10	33279631	33 279 631.0	33 279 631	33 279 638.2
3	300	480	10	33248818	33 248 818.0	33 248 818	33 248 818.0
4	300	480	10	33211353	33 211 353.0	33 211 353	33 211 354.2
5	300	480	10	33234319	33 234 319.0	33 234 319	33 234 319.0

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Table 2: Aggregated main results for uniform instances.

index	m	n	k	LB	SA	Gurobi	HG2
1	300	1020	10	71844251	71 844 251.0	71 844 251	71 844 251.0
2	300	1020	10	71891931	71 891 931.0	71 891 931	71 891 936.0
3	300	1020	10	71900182	71 900 182.0	71 900 182	71 900 182.0
4	300	1020	10	71825623	71 825 623.0	71 825 623	71 825 623.0
5	300	1020	10	71843983	71 843 983.0	71 843 983	71 843 983.0
1	300	2040	10	145227714	145 227 714.0	145 227 714	145 315 447.9
2	300	2040	10	145286523	145 286 523.0	145 286 523	145 367 278.7
3	300	2040	10	145108246	145 108 246.0	145 108 246	145 205 498.2
4	300	2040	10	145203742	145 203 742.0	145 203 742	145 292 409.1
5	300	2040	10	145182431	145 182 431.0	145 182 431	145 272 401.0